



Vendor Performance Analysis

By Parshv Patel

Data Description

Datasets Used:

- `begin_inventory` – 206k records of starting stock quantities
- `end_inventory` – 224k records of closing stock
- `purchases` – 2.37M transactions (vendor, brand, purchase price, units, dollars)
- `purchase_prices` – 12k records of brand-wise benchmark/actual product prices
- `sales` – 12.8M sales transactions (quantity, sales dollars, price, tax, revenue)
- `vendor_invoice` – 5.5k vendor invoices (PO numbers, dollar amounts, freight)

Worked with a total of 15.65M combined data rows from the 6 CSV files.

Purpose of Dataset:

- Track purchases, sales, and inventory at the vendor-brand level
- Integrate pricing, freight, and invoices to evaluate vendor efficiency and profitability
- Provide foundation for vendor selection, product pricing strategies, and contract negotiations

Business Problem

Effective inventory and sales management are critical for optimizing profitability in the retail and wholesale industry. Companies need to ensure that they are not incurring losses to inefficient pricing, poor inventory turnover, or vendor dependency. The goal of this analysis is to:

- Identifying underperforming brands that require promotional or pricing adjustments
- Determining top vendors contributing to sales and gross profit.
- Analyze the impact of bulk purchasing to sales and gross profit.
- Assess inventory turnover to reduce holding costs and improve efficiency.
- Investigate the profitability variance between high-performing and low-performing vendors.

Data Cleaning & Preparation

Data Cleaning Steps:

- Removed rows with invalid purchase prices (`PurchasePrice > 0`) to ensure accuracy
- Filled 178 null values in Sales (Quantity, Dollars, Price, Excise Tax) with `0` since vendors purchased items but recorded no sales
- Trimmed whitespace errors from `VendorName` using `.strip()` for consistency
- Ensured column data types were appropriate for numeric and categorical processing

Query Optimization:

- Initial query combining 4 large tables ran >9 minutes, unsuitable for near real-time analytics
- Re-engineered using CTEs and aggregations → reduced runtime to 44 seconds
- Combined results into `vendor_sales` summary table (10,692 rows, 13 features)

Feature Engineering

New Metrics Created for Analysis:

- Gross Profit = Sales Dollars – Purchase Dollars
- Profit Margin (%) = $\text{Gross Profit} \div \text{Total Sales} \times 100$
- Stock Turnover = $\text{Sales Quantity} \div \text{Purchase Quantity}$
 - = 1 → items purchased were sold out
 - > 1 → *vendor sold existing stock in addition to new purchases*
- Sales-to-Purchase Ratio for performance benchmarking

Final Dataset:

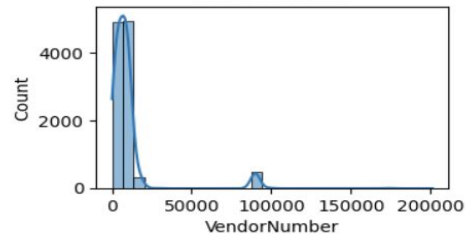
- Created `vendor_sales_summary` table with 17 columns and 10,692 rows
- Stored back into SQL DB with optimized schema for fast retrieval, safe storage, and integration into dashboards
- Supports real-time analytics for vendor performance monitoring and profitability insights

Exploratory Data Analysis Insights

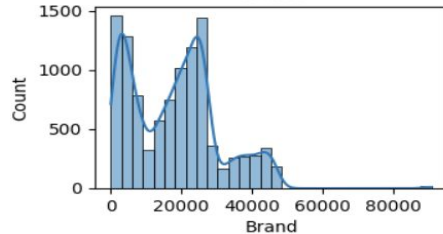
Summary Statistics

	count	mean	std	min	25%	50%	75%	max
VendorNumber	10692.0	10650.649458	18753.519148	2.000000	3951.000000	7153.000000	9552.000000	2.013590e+05
Brand	10692.0	18039.228769	12662.187074	58.000000	5793.500000	18761.500000	25514.250000	9.063100e+04
AvgActualPrice	10692.0	35.643671	148.246016	0.490000	10.990000	15.990000	28.990000	7.499990e+03
AvgVolume	10692.0	847.360550	664.309212	50.000000	750.000000	750.000000	750.000000	2.000000e+04
TotalPurchaseQuantity	10692.0	3140.886831	11095.086769	1.000000	36.000000	262.000000	1975.750000	3.376600e+05
TotalPurchaseDollars	10692.0	30106.693372	123067.799627	0.710000	453.457500	3655.465000	20738.245000	3.811252e+06
TotalSalesQuantity	10692.0	3077.482136	10952.851391	0.000000	33.000000	261.000000	1929.250000	3.349390e+05
TotalSalesDollars	10692.0	42239.074419	167655.265984	0.000000	729.220000	5298.045000	28396.915000	5.101920e+06
TotalSalesPrice	10692.0	18793.783627	44952.773386	0.000000	289.710000	2857.800000	16059.562500	6.728193e+05
TotalExciseTax	10692.0	1774.226259	10975.582240	0.000000	4.800000	46.570000	418.650000	3.682428e+05
FreightCost	10692.0	61433.763214	60938.458032	0.090000	14069.870000	50293.620000	79528.990000	2.570321e+05
GrossProfit	10692.0	12132.381048	46224.337964	-52002.780000	52.920000	1399.640000	8660.200000	1.290668e+06
ProfitMargin	10514.0	-15.885227	447.289882	-23730.638953	15.353839	30.778375	40.210967	9.971666e+01
StockTurnover	10692.0	1.706793	6.020460	0.000000	0.807229	0.981529	1.039342	2.745000e+02
SalestoPurchaseRatio	10692.0	2.504390	8.459067	0.000000	1.153729	1.436894	1.665449	3.529286e+02

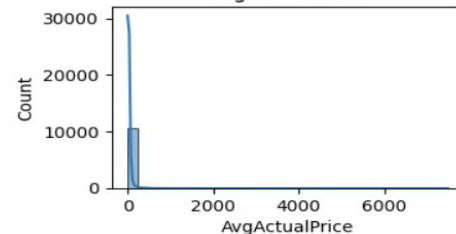
VendorNumber



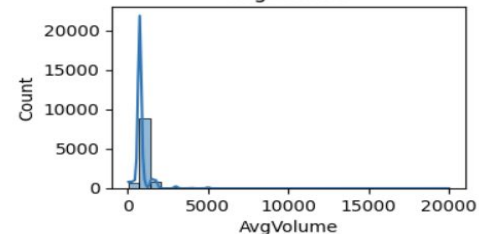
Brand



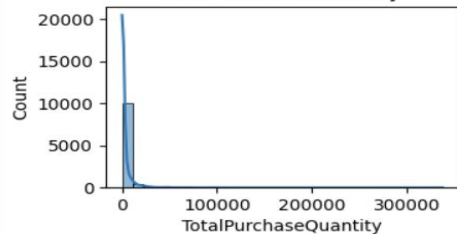
AvgActualPrice



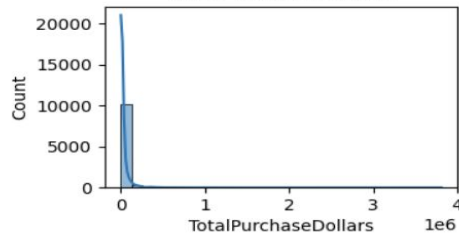
AvgVolume



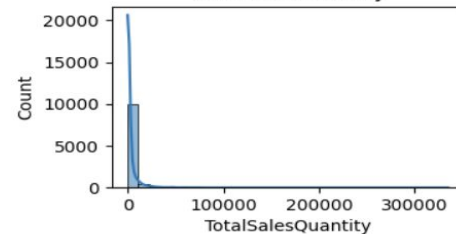
TotalPurchaseQuantity



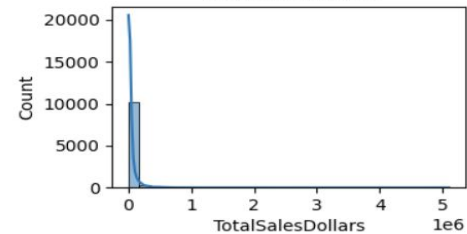
TotalPurchaseDollars



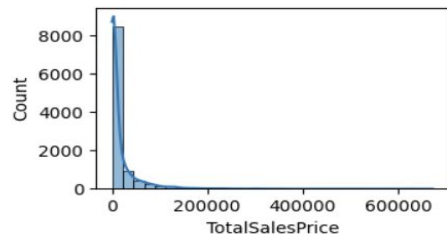
TotalSalesQuantity



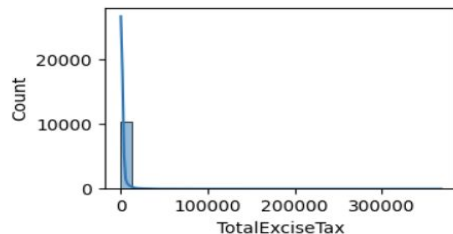
TotalSalesDollars



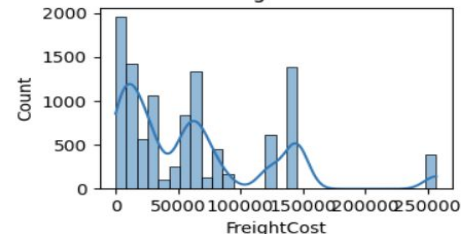
TotalSalesPrice



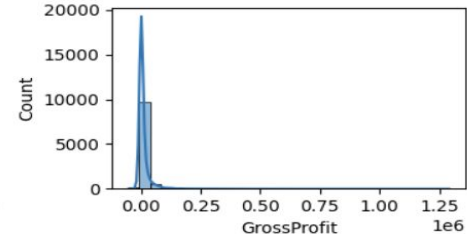
TotalExciseTax



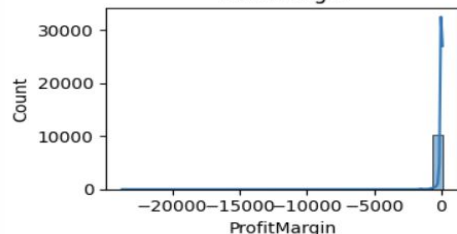
FreightCost



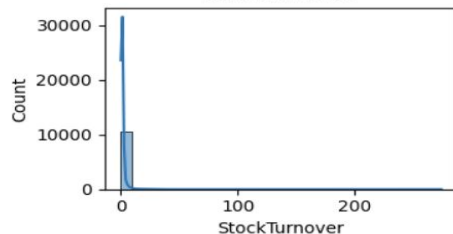
GrossProfit



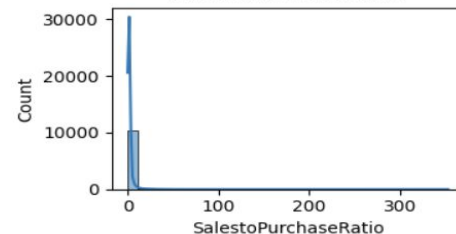
ProfitMargin

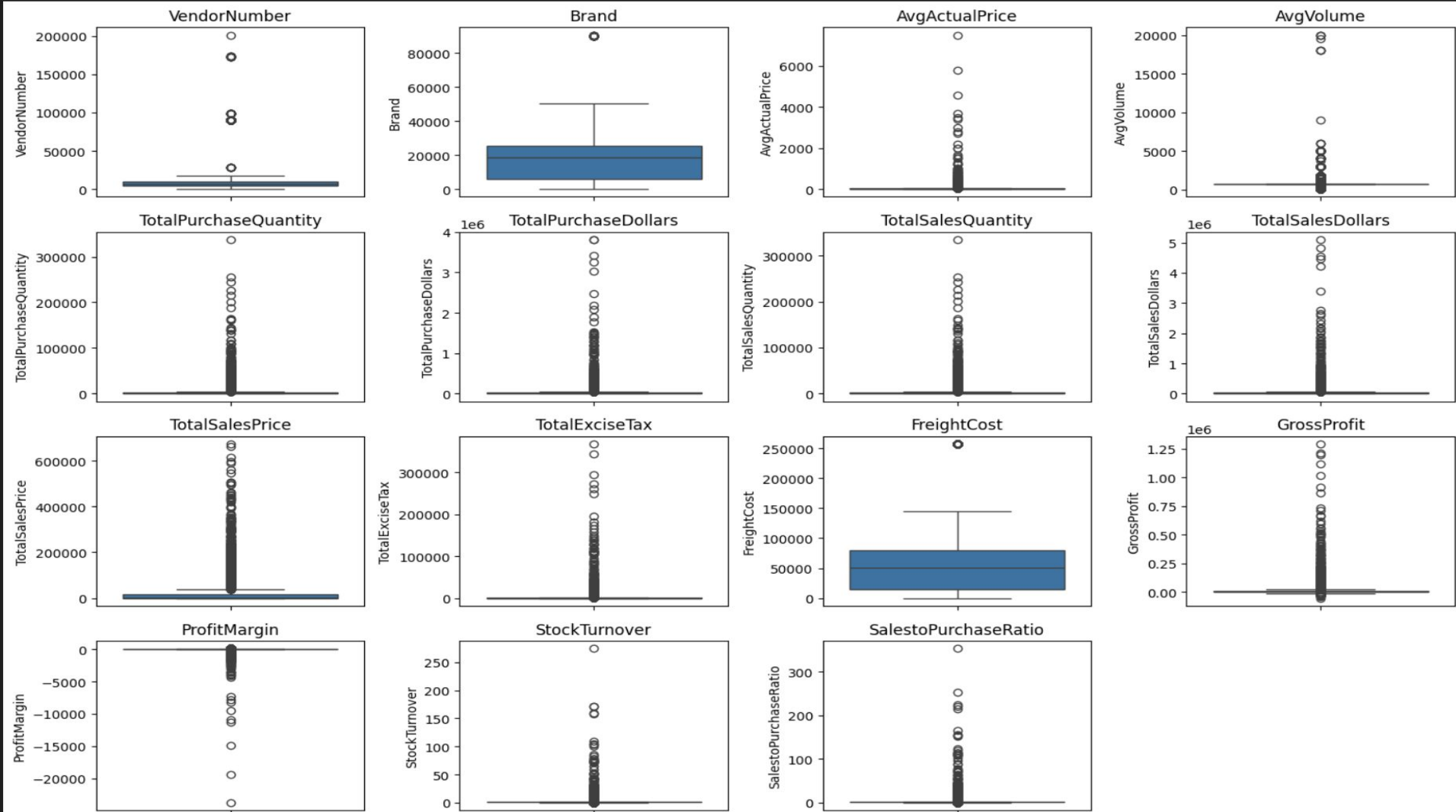


StockTurnover



SalestoPurchaseRatio





Gross Profit: Minimum of -52,003, indicating potential losses due to high costs or heavy discounts. This could be due to selling products at lower prices than their purchase costs.

Profit Margin: Minimum value of -23,730, which suggest cases where revenue is even lower than total costs, leading to extreme negative profit margins.

Total Sales Quantity and Sales Dollars: Minimum values of 0, meaning some products were purchased but never sold. These could be due to slow-moving or obsolete stock, leading to inventory inefficiencies.

Actual Price: Max value of 7500 which is significantly higher than the mean 35.64 and a high standard deviation of 148, indicating potential premium products.

Freight Cost: Huge variation from 0.09 to 257,032, suggesting logistics inefficiencies or bulk shipments, or inconsistent shipping costs across different products.

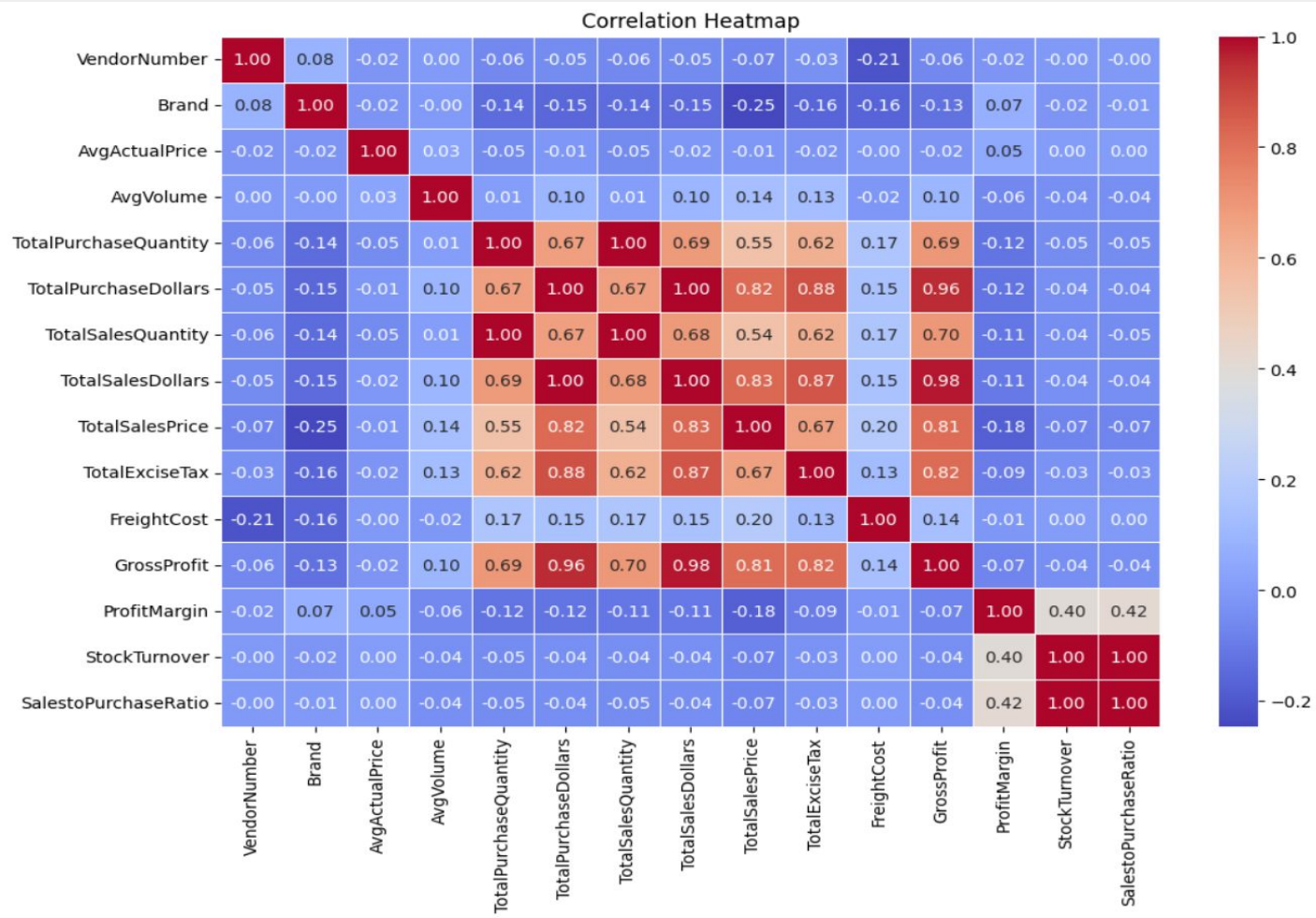
Stock Turnover: Ranges from 0 to 274.5, implying some products sell extremely fast while others remain in stock indefinitely. Value more than 1 indicates that Sold quantity for that product is higher than purchased quantity due to sales being fulfilled from older stock.

Data Filtering

To enhance the reliability of the insights, I removed inconsistent data points where:

- Gross Profit ≤ 0 (to exclude transactions leading to losses)
- Profit Margin ≤ 0 (to ensure analysis focuses on profitable transactions)
- Total Sales Quantity = 0 (to eliminate inventory that was never sold)

Correlation Insights



- **Purchase Price** has weak correlations with **Total Sales Dollars** (-0.012) and **Gross Profit** (-0.016), suggesting that price variations do not significantly impact sales revenue or profit.
- Strong correlation between **total purchase quantity** and **total sales quantity** (1.00) which confirms efficient inventory turnover.
- Negative correlation between **profit margin** and **total sales price** (-0.18) suggests that as sales price increases, margins decrease, possibly due to competitive pricing pressures.
- **Stock Turnover** has weak negative correlations with **Gross Profit** (-0.04) and moderate positive correlation with **Profit Margin** (0.40), indicating the faster turnover does not necessarily result in higher profitability.

Research Question and Key Findings

1) Brands for Promotional or Price Adjustments

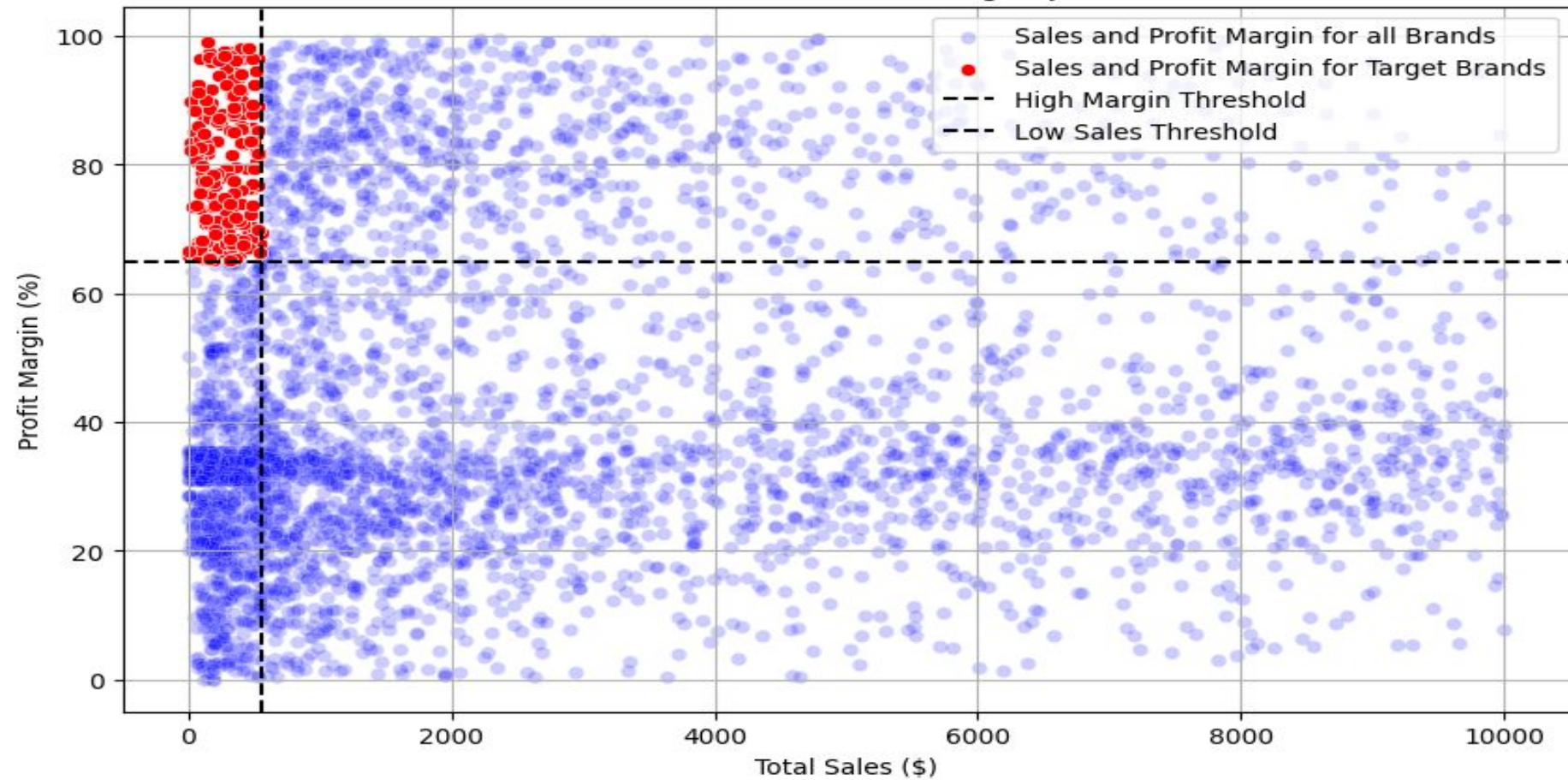
198 brands exhibit lower sales but higher profit margins, which could benefit from targeted marketing. Promotions, or price optimizations to increase volume without compromising profitability.

Brands with low Sales but high Profit Margins are:

	Description	TotalSalesDollars	ProfitMargin
0	Santa Rita Organic Svgn Bl	9.99	66.466466
1	Debauchery Pnt Nr	11.58	65.975820
2	Concannon Glen Ellen Wh Zin	15.95	83.448276
3	Crown Royal Apple	27.86	89.806174
4	Sauza Sprklg Wild Berry Marg	27.96	82.153076
...	...	...	...
193	Nanbu Bijin Southern Beauty	535.68	76.747312
194	Dad's Hat Rye Whiskey	538.89	81.851584
195	A Bichot Clos Marechaudes	539.94	67.740860
196	Sbragia Home Ranch Merlot	549.75	66.444748
197	Goulee Cos d'Estournel 10	558.87	69.434752

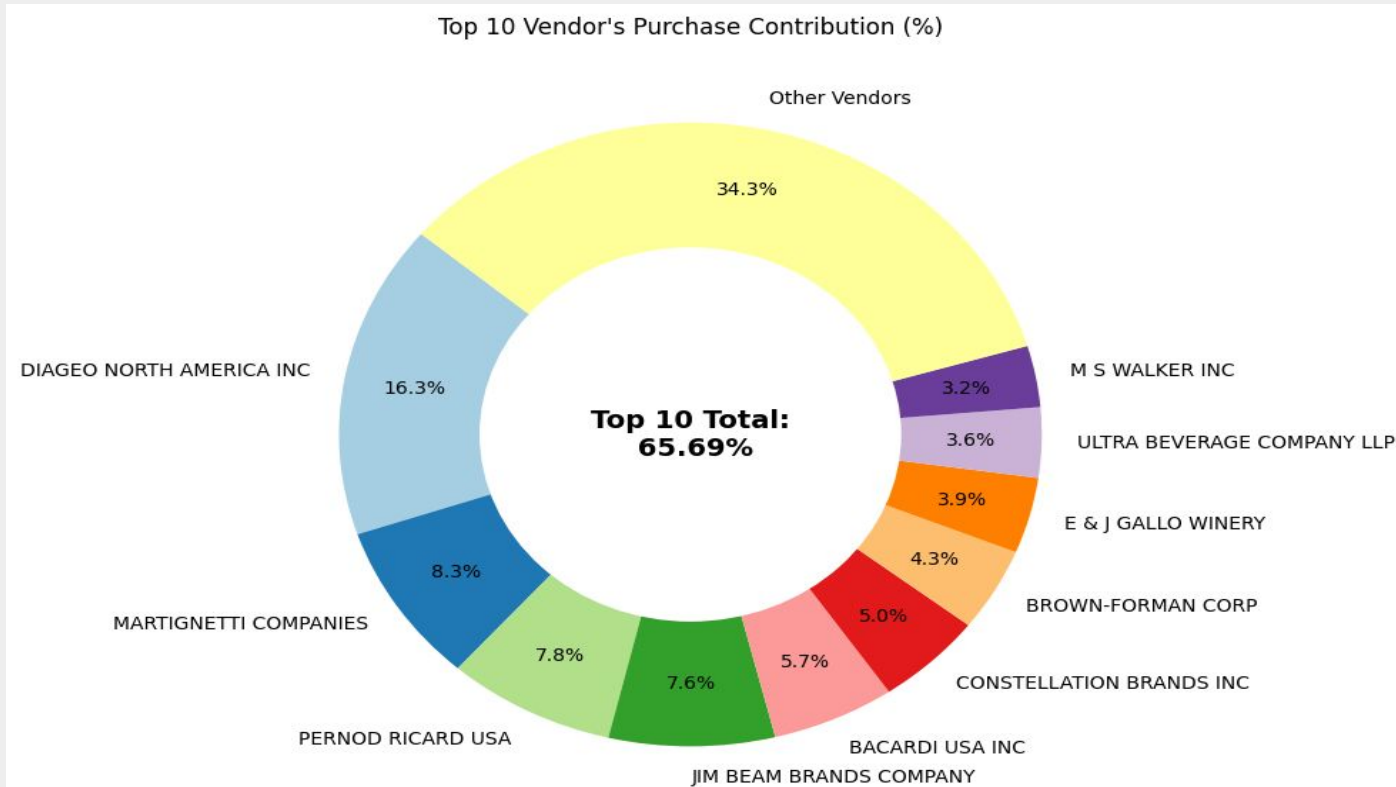
198 rows × 3 columns

Brands for Promotional or Pricing Adjustments

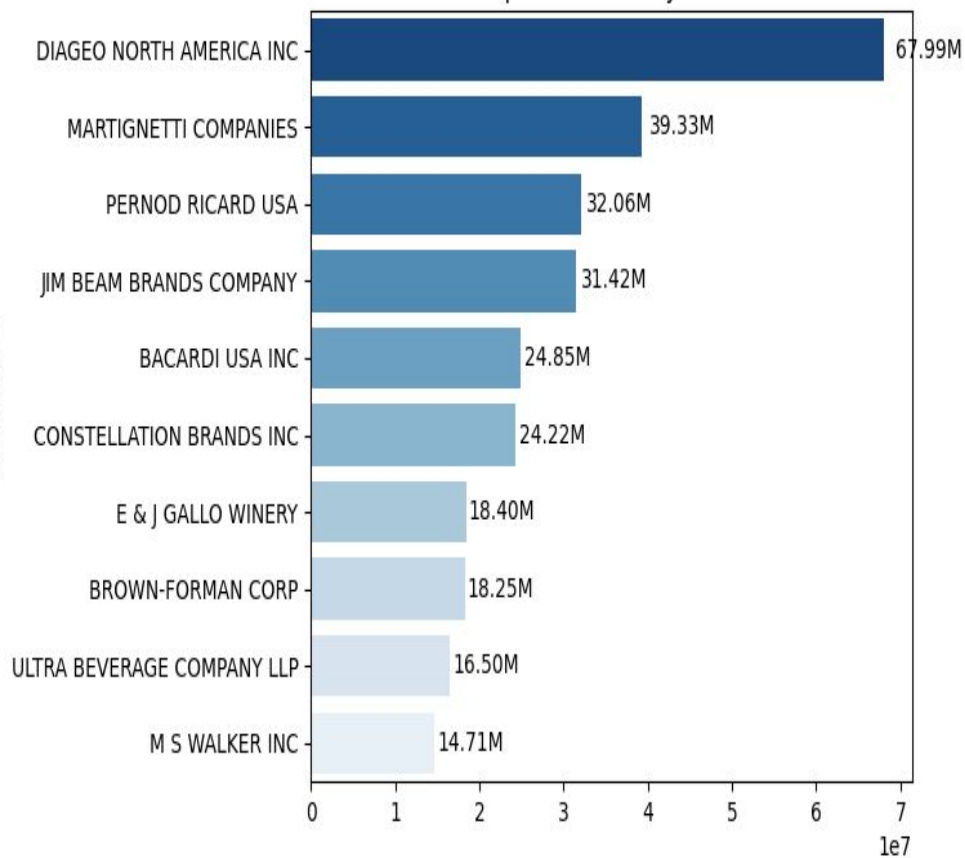


2) Top Vendors by Sales and Purchase Contribution

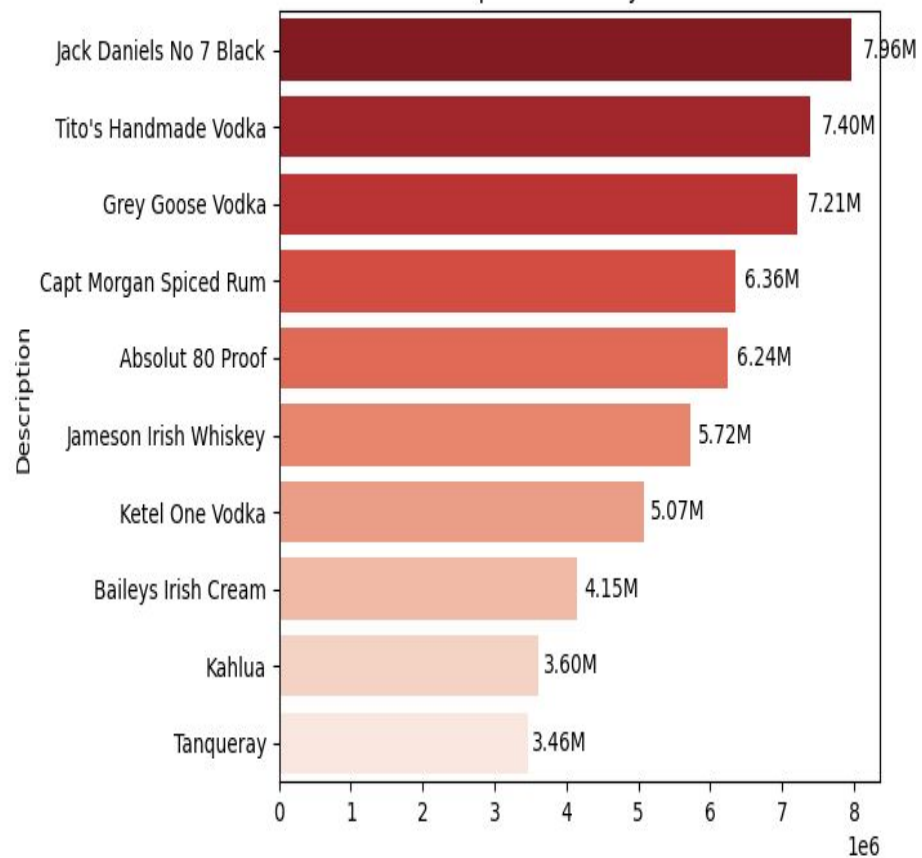
The top 10 vendors contributed 65.7% of total purchases, while the remaining 109 vendors contribute only 34.3%. This over-reliance on a few vendors may introduce risks such as supply chain disruptions, indicating a need for diversification.



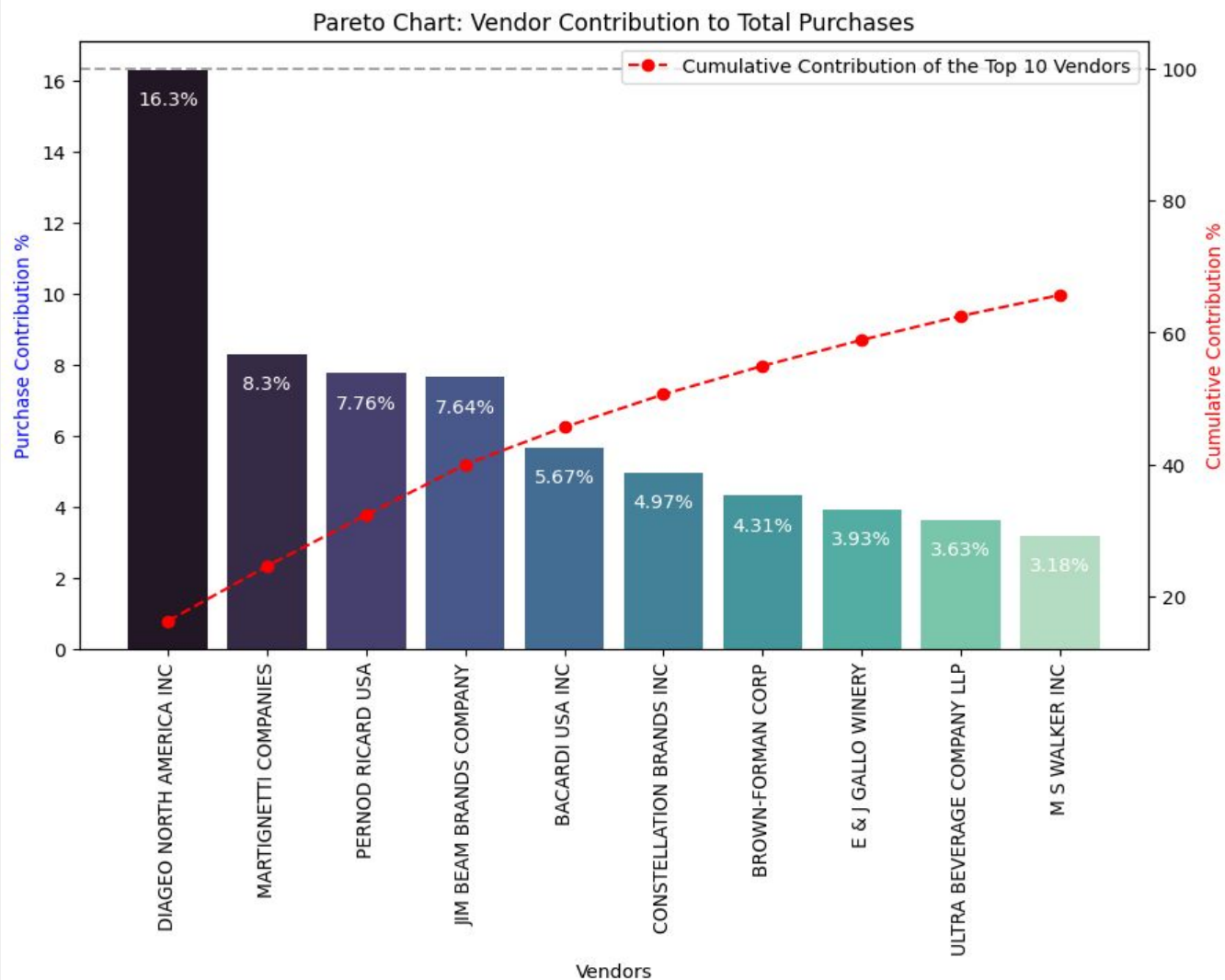
Top 10 Vendors by Sales



Top 10 Brands by Sales



This Pareto Chart displays data in a bar graph format, sorted from highest to lowest frequency, and a cumulative line to highlight that just the top 10 vendors purchase 65.7% of products from the 119 total vendors



3) Impact of Bulk Purchasing on Cost Savings

Vendors buying in large quantities receive a 72% lower unit cost (\$10.78 per unit for large orders vs high \$39.06 unit costs in smaller orders).

Bulk pricing strategies encourage larger orders, increasing total sales while maintaining profitability.

UnitPurchasePrice	
OrderSize	
Small	39.057543
Medium	15.486414
Large	10.777625

4) Identifying Vendors with Low Inventory Turnover

Total Unsold Inventory Capital: \$3.71M

Slow-moving inventory increases storage costs, reduce cash flow efficiency, and affects overall profitability. Identifying vendors with low inventory turnover enables better stock management, minimizing financial strain.

VendorName	StockTurnover
ALISA CARR BEVERAGES	0.615385
HIGHLAND WINE MERCHANTS LLC	0.708333
PARK STREET IMPORTS LLC	0.751306
Circa Wines	0.755676
Dunn Wine Brokers	0.766022
CENTEUR IMPORTS LLC	0.773953
SMOKY QUARTZ DISTILLERY LLC	0.783835
TAMWORTH DISTILLING	0.797078
THE IMPORTED GRAPE LLC	0.807569
WALPOLE MTN VIEW WINERY	0.820548

VendorName	UnsoldInventoryValue
DIAGEO NORTH AMERICA INC	984.23K
JIM BEAM BRANDS COMPANY	761.21K
PERNOD RICARD USA	647.40K
WILLIAM GRANT & SONS INC	538.71K
E & J GALLO WINERY	369.73K
SAZERAC CO INC	273.37K
BROWN-FORMAN CORP	247.06K
CONSTELLATION BRANDS INC	227.37K
MOET HENNESSY USA INC	197.64K
MAJESTIC FINE WINES	180.73K

5) Profit Margin Comparison: High vs Low-Performing Vendors

Top Vendors' Profit Margin (95% CI): (30.74%, 31.61%) Mean: 31.17%

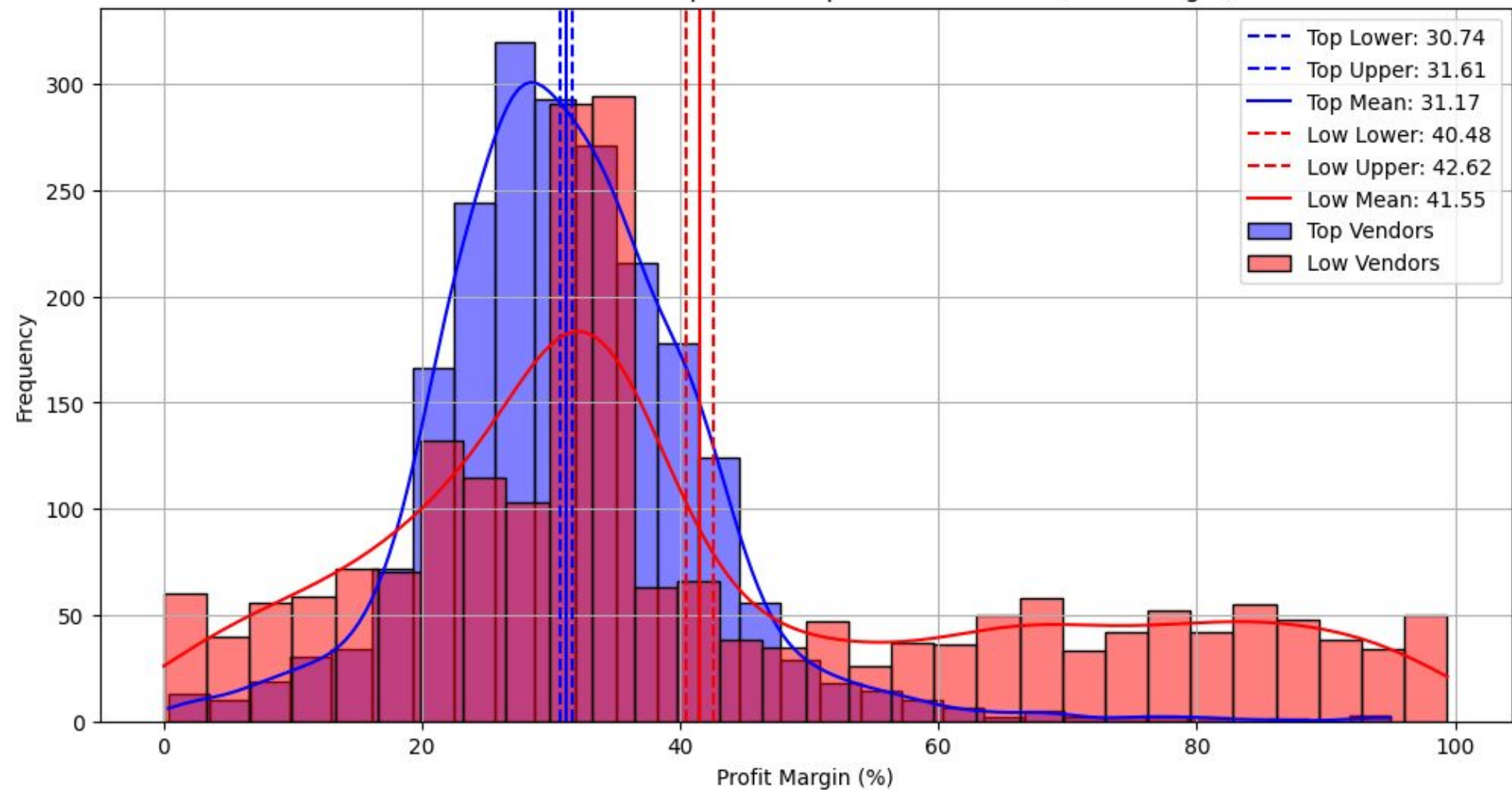
Low Vendors' Profit Margin (95% CI): (40.48%, 42.62%) Mean: 41.55%

Low-performing vendors maintain higher margins but struggle with sales volumes, indicating potential pricing inefficiencies or market reach issues.

Actionable Insights:

- Top-performing vendors: Optimize profitability by adjusting pricing, reducing operational costs, or offering bundled promotions.
- Low-performing vendors: Improve marketing

Confidence Interval Comparison: Top vs Low Vendors (Profit Margin)



Hypothesis Testing:

- **H0 (Null Hypothesis):** There is no significant difference in the mean profit margins of top-performing and low-performing vendors. Any observed difference in the samples are random and due to chance.
- **H1 (Alternative Hypothesis):** The mean profit margins of top-performing and low-performing vendors are significantly different. Any observed difference in the samples are due to factors other than chance.
- α : 0.05

Result: T-Statistic :-17.644, P-value: 0.0

We reject the H0. There is statistically significant evidence to support that there is a difference in profit margins between top and low-performing vendors. The data is consistent with Alternative Hypothesis. Even if Null Hypothesis is True, there is 5% chance of falsely rejecting the Null Hypothesis which is very low by chance.

Implication: High-margin vendors may benefit from better pricing strategies, while top-selling vendors could focus on cost efficiency.

Final Recommendations

- Re-evaluate pricing for low-sales, high-margin brands to boost sales volume without sacrificing profitability.
- Diversify vendor partnerships to reduce dependency on a few suppliers and mitigate supply chain risks.
- Leverage bulk purchasing advantages to maintain competitive pricing while optimizing inventory management.
- Optimize slow-moving inventory by adjusting purchase quantities, launching clearance sales, or revising storing strategies.
- Enhance marketing and distribution strategies for low-performing vendors to drive higher sales volumes without compromising profit margins.
- By implementing these recommendation, the company can achieve sustainable profitability, mitigate risks, and enhance overall operational efficiency.

Vendor Performance Dashboard

441.41M

Total Sales (\$)

307.34M

Total Purchase (\$)

134.07M

Gross Profit (\$)

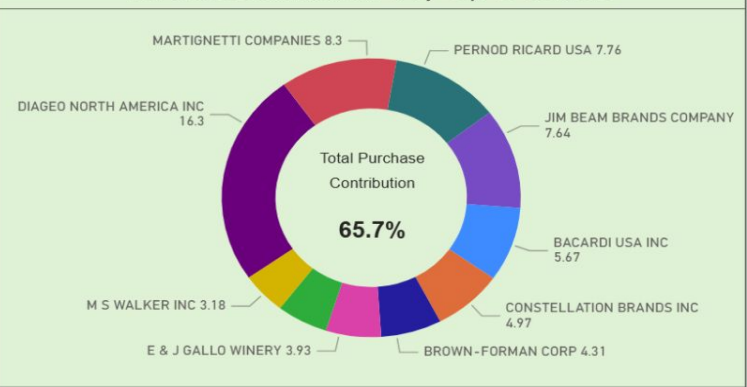
38.72

Profit Margin (%)

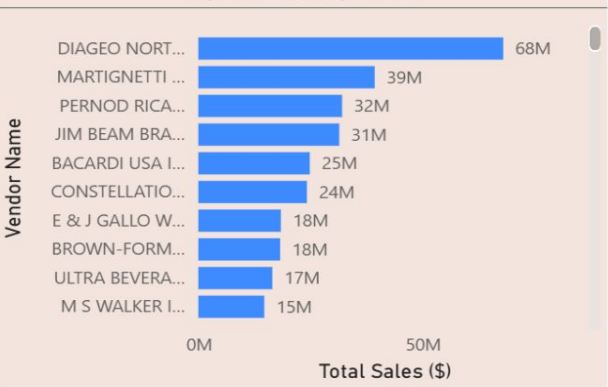
3.71M

Unsold Capital (\$)

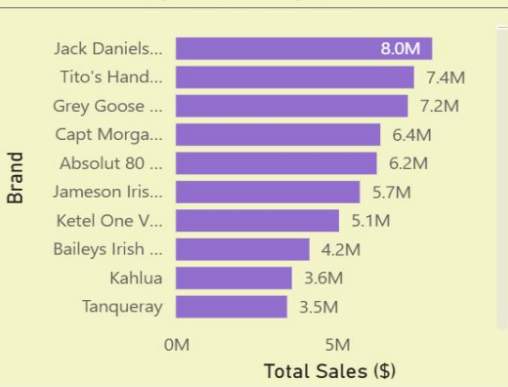
Purchase Contribution % by Top 10 Vendors



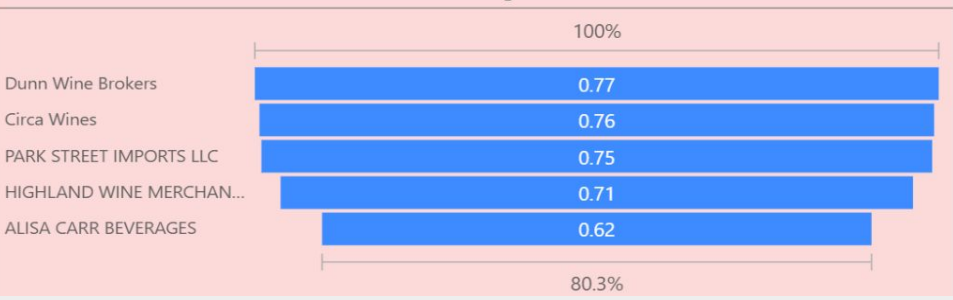
Top Vendors by Sales



Top Vendors by Sales



Low Performing Vendors



Low Performing Brands

